Writing Method of Moments code for Inverse Source Reconstruction Algorithm

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Project Report

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I. Introduction

Large parabolic reflector antennas used by ISRO play a critical role in satellite communication, deep-space telemetry, and tracking operations. The performance of these antennas depends heavily on the precise alignment and integrity of every individual panel on the reflector surface. Even small structural deviations, surface defects, or alignment errors on a single panel can significantly distort the radiation pattern, reduce gain, or introduce unexpected sidelobes. Diagnosing such issues is currently a labor-intensive process, often requiring manual inspection of each panel, which is time-consuming, costly, and difficult for large apertures installed at operational sites.

To address this challenge, ISRO is exploring the use of drones attached with antennas to measure the near-field electric field around the antenna while it is operating. By processing this measured field, it is possible to recover the surface current distribution on the reflector and reconstruct the far-field radiation pattern, enabling engineers to identify which specific region of the parabolic surface is degrading performance. This approach provides a non-contact, automated method of diagnosing reflector health, eliminating the need for commercial electromagnetic simulation tools and allowing fully in-house processing tailored to ISRO's requirements.

This project focuses on developing a Python-based Method of Moments (MoM) solver capable of computing surface currents, near-field and far-field quantities directly from measured data, and evaluating the condition of individual reflector panels. The aim is to create a reliable, open source, and customizable electromagnetic analysis pipeline that can be integrated into ISRO's drone-based antenna inspection workflow.

II. Method of Moments

In this section, the progress made in implementation of Method of Moments code after last semester is discussed.

A. Performance Improvement

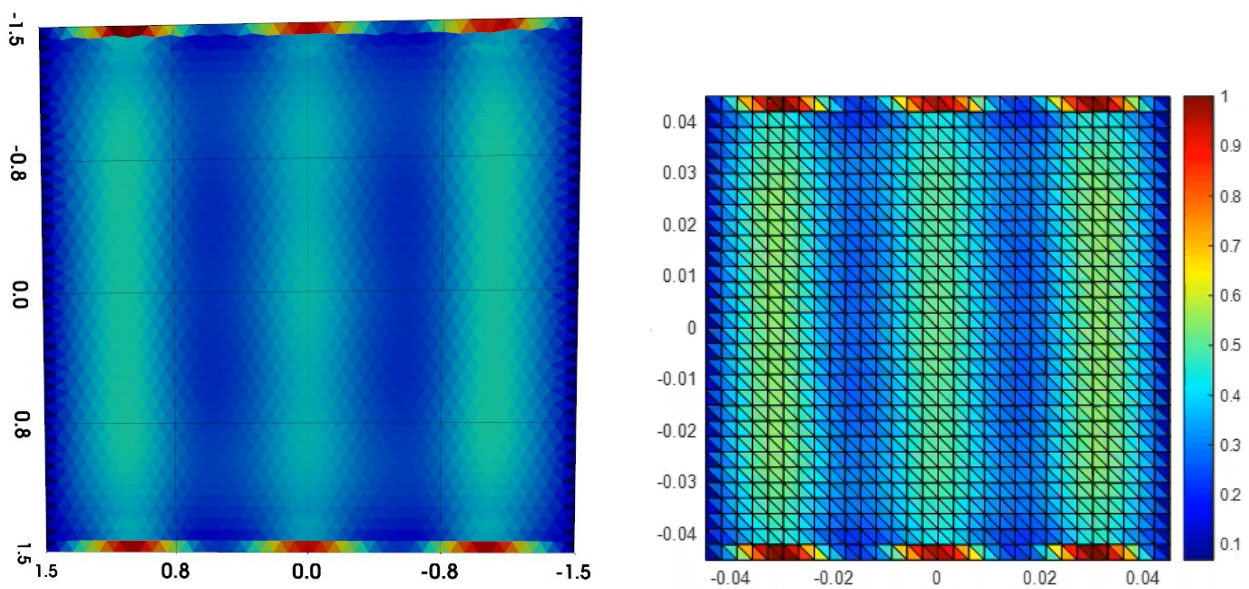
Previously the Method of Moments code was implemented using numpy which had terrible runtime. We were able to further improve the runtime using numpy vectorization but the code was still taking hours. We then shifted to completely rewriting our code in Taichi which is a domain-specific language embedded in Python that helps you write high-performance parallel programs. We saw major improvement in runtime as you can see in Table I.

Implementation Method	Gauss Quadrature Rule	Runtime
Numpy	7-point	8:00:00
Numpy with Vectorization	7-point	1:30:00
Numpy with Vectorization	16-point	2:15:00
Taichi	7-point	0:00:35
Taichi	16-point	0:01:40

TABLE I: Comparison of Runtime Performance across Different Implementation Methods and Quadrature Rules

B. Results

Previously the code was giving different current distribution for different meshes. Finely meshed and coarsely meshed surfaces both gave different but incorrect results. If we changed the meshing function which decides the qualities of the triangles of the mesh we again got different result. All that is fixed now and the obtained current distribution is an exact match to the expected current distribution as can be see from Figure 1.



(a) Obtained current distribution

(b) Expected current distribution

Fig. 1: Comparison between obtained and expected current distributions on the antenna surface.

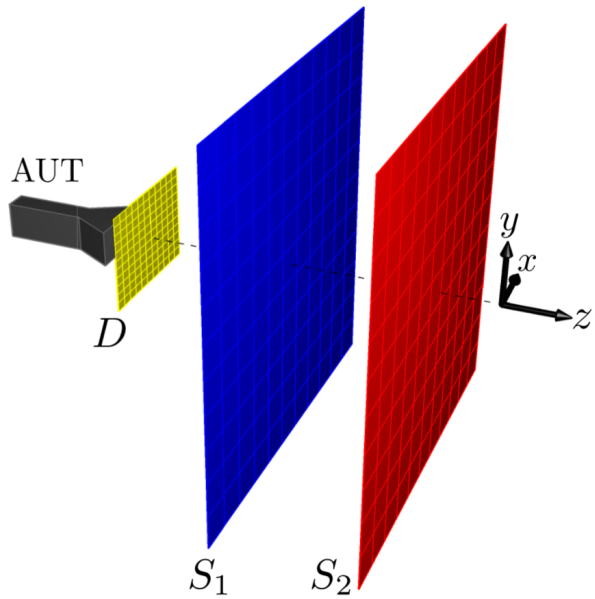


Fig. 2: Planar Near Field measurement setup for MR-SRM

C. Usecases

We won't be using this code practically since we realized that it needs phase and magnitude of the electric field data and also the electric field data needs to be collected very close to the surface of the antenna which will be difficult for the drone. But the implementation of this code helped me learn the power of Taichi of runtime improvement and also how to handle singularity issues when integrating. Also this code is not based on iteration and convergence and its memory requirement is low too.

III. MR-SRM Solver

A. Working

The Multiplicatively Regularized Source Reconstruction Method (MR-SRM) [1] works by reconstructing an equivalent current distribution that radiates the same fields as the antenna under test (AUT). This is achieved by minimizing a nonlinear cost functional that relates measured amplitude-only data to fields produced by unknown currents.

1) Forward Problem Formulation: The algorithm relates the unknown equivalent magnetic currents ($\vec{M}$) on a reconstruction domain (D) to the radiated electric field ($\vec{E}$) via the electro-

magnetic equivalence principle. For planar measurement setups, the equations decouple into two scalar integral forms:

$$E_x(\vec{r}) = 2\int_D M_y(\vec{r}')\frac{\partial g_0(\vec{r},\vec{r}')}{\partial z'}d\vec{r}' \tag{1}$$

$$E_y(\vec{r}) = -2\int_D M_x(\vec{r}')\frac{\partial g_0(\vec{r},\vec{r}')}{\partial z'}d\vec{r}' \tag{2}$$

where g_0 is the free space Green's function:

$$g_0(\vec{r},\vec{r}') = \frac{e^{-jk_0|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} \tag{3}$$

By discretizing the domain D into pulse basis functions, the system is represented as a matrix mapping:

$$A_i u = f_i \tag{4}$$

where u represents the coefficients of the equivalent currents and f_i is the electric field produced on measurement planes S_1 and S_2.

2) Multiplicatively Regularized Cost Functional: To solve the non-linear inverse problem without phase data, the MR-SRM minimizes a cost functional $\mathcal{C}(u)$ comprised of data misfit and regularization terms:

$$\mathcal{C}(u) = (\mathcal{C}_1(u) + \mathcal{C}_2(u))\mathcal{C}_{MR}(u) \tag{5}$$

3) Data Misfit Terms: The terms $\mathcal{C}_1$ and $\mathcal{C}_2$ quantify the discrepancy between the measured field amplitudes and those produced by the current estimate on two distinct planes:

$$\mathcal{C}_i(u) = \eta_i||A_i u|^2 - |f_i|^2||^2_{S_i} \tag{6}$$

where η_i is a normalization factor based on the measured data.

4) Multiplicative Regularizer: The regularizer $\mathcal{C}_{MR}$ is a weighted L_2-norm designed to stabilize the ill-posed inversion:

$$\mathcal{C}_{MR}(u) = \int_D b^2(\vec{r}')(|\nabla u(\vec{r}')|^2 + \delta^2)d\vec{r}' \tag{7}$$

Because the regularizer is multiplied by the data misfit, the weighting is adaptive: a high data error leads to stronger regularization, while a low error allows the algorithm to resolve finer details.

5) Optimization via Conjugate Gradient: The functional is minimized using the Conjugate Gradient (CG) algorithm. At each iteration n, the solution is updated along a search direction v_n:

$$u_{n+1} = u_n + \alpha_n v_n \tag{8}$$

$$v_n = g_n + \beta_n v_{n-1} \tag{9}$$

The gradient g_n is calculated using a closed-form expression:

$$g_n = 2\eta_1 A_1^H r_{1,n} \odot A_1 u_n + 2\eta_2 A_2^H r_{2,n} \odot A_2 u_n + (\mathcal{C}_1(u_n) + \mathcal{C}_2(u_n)) g_n^{MR} \tag{10}$$

where $\odot$ denotes the Hadamard product and $g_n^{MR} = -\nabla \cdot (b_n^2 \nabla u_n)$.

6) Adaptive Regularization Mechanism: The steering parameter δ_n adaptively controls the regularization behavior:

- **Initial Stage**: When the data error is high, δ_n is large, causing the operator to act as a Laplacian smoother to provide stability.
- **Convergence Stage**: As the error decreases, δ_n becomes small, enabling edge-preserving characteristics that allow the algorithm to accurately reconstruct physical discontinuities and defects in the current distribution.

B. Runtime Improvements

In the MR-SRM algorithm we used Taichi Lang to build the A matrices for both the measurement surfaces for fast execution. Unfortunately the rest of the code had to be implemented using numpy since Taichi Lang does not come with built-in functions like gradient calculation of matrices, calculating hermitian matrices, matrix multiplication and many other operations that are required by the algorithm. It is possible to do this but currently the code's runtime is acceptable at around 1 minute so any improvement made by Taichi Lang is not worth the effort.

C. Results

The paper we used doesn't discuss the size of the measurement planes, samples at which data should be taken, distance between measurement planes, distance of measurement planes from the antenna and how finely we need to mesh the reconstruction surface for accurate reconstruction.

We can't sample collect electric field data at very small intervals since that leads to huge datasets and long runtime and at large intervals the reconstruction might not be accurate.

We also need to see if a very coarse meshing of reconstruction surface leads to inaccurate reconstruction or just poor resolution. We also tested if the ζ value used in the first initial guess actually affects convergence to accurate reconstruction or not and it did not. So its best to keep ζ value to be 1 and save some runtime.

For testing the code we simulated a horn parabolic antenna in CST software and collected data at different distances and different samples from the parabolic dish. The current is being reconstructed on the parabolic dish. The wavelength λ of the electric field is 15mm.

The terminology used in the image captions are: Dx $\rightarrow$ distance between each sample point on the measurement plane, Size $\rightarrow$ size of the measurement planes S1 and S2, mesh-length $\rightarrow$ length of each quadrilateral element of the mesh for surface D on which current is being reconstructed, S1-S2 $\rightarrow$ distance of measurement planes S1 and S2 from the antenna rescpectively. All the values are in millimeters.

The most accurate results we get are from Fig 9 and Fig 10 followed by Fig 4. From this we can first infer that we can keep Dx at around λ without suffering from inaccuracies in current reconstruction. Second we notice moving away we get a more circular reconstruction which is closer to the actual current. From Fig 6 we can see this effect even more clearly that moving closer is giving us worse reconstruction but this maybe due to the fact that the horn part of the antenna is itself at 120mm distance from the dish. From Fig 7 we see a coarse mesh leads to poor resolution but not an inaccurate reconstruction so if we are constraint by runtime or memory its fine to make the mess more coarse. In Fig 8 we reduce the size of the measurement planes S1 and S2 but also sample the data more closely but this gives a very inaccurate reconstruction.

IV. Future Works

The drone's antenna that we'll be using for collecting the electric field data at different points won't give the actual electric field values. We must integrate the raw data with antenna's weighting function to get the actual values which can be done using Gaussian Quadrature implemented using Taichi for fast performance.

V. Conclusion

The code written to reconstruct the current present on an antenna's surface is working, optimized for runtime and tested for different configurations of measurement and reconstruction

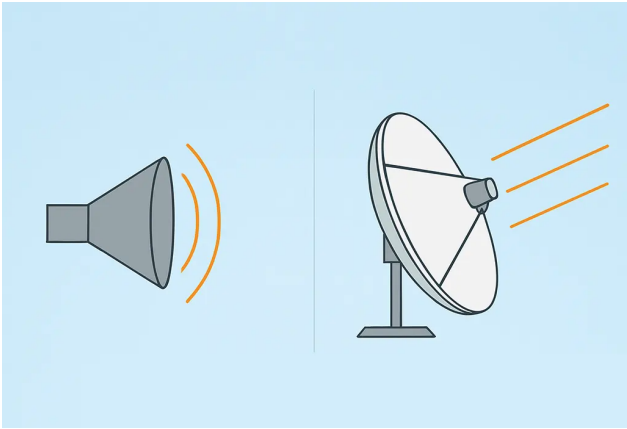


Fig. 3: Horn Parabolic antenna

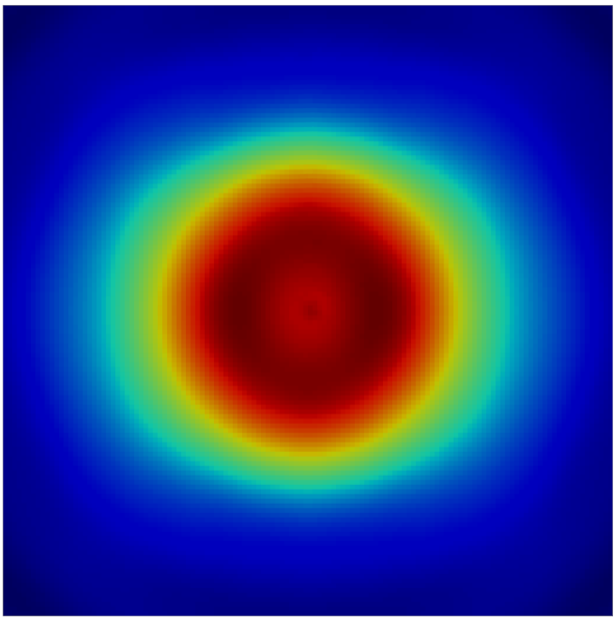


Fig. 4: Dx: 7.5, Size: 780 X 780, mesh-length: 1.27, S1-S2: 220 and 250

planes for the user to decide what's best for their purpose. The best configuration is to keep the measurement plane's size between 30λ X 30λ and 50λ X 50λ, mesh-length at about λ/5 and distance of the measurement planes from the antenna at about 20λ. You can checkout this Github repository containing all the codes related to this project.

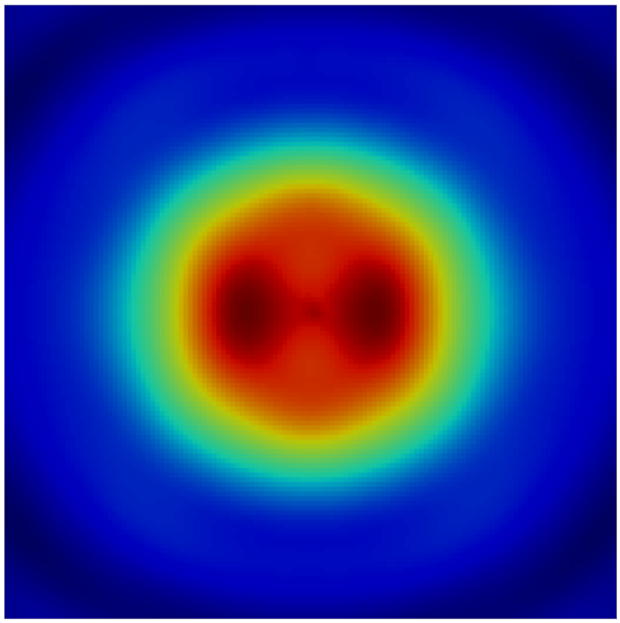


Fig. 5: Dx: 7.5, Size: 780 X 780, mesh-length: 1.27, S1-S2: 180 and 250

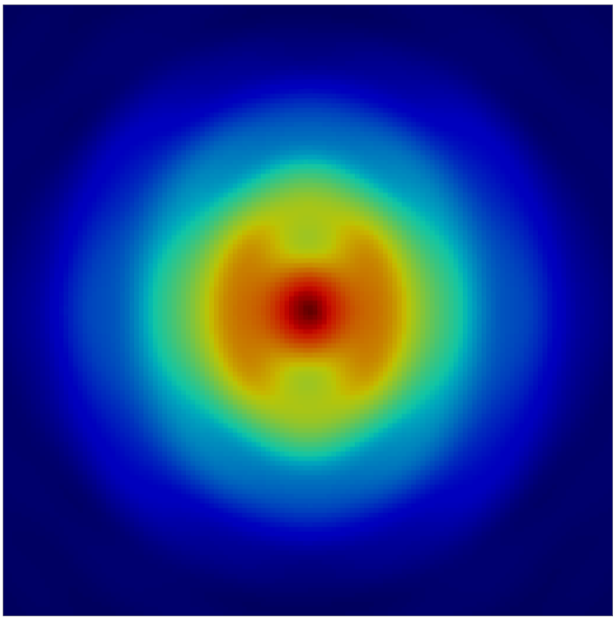


Fig. 6: Dx: 7.5, Size: 780 X 780, mesh-length: 1.27, S1-S2: 120 and 150

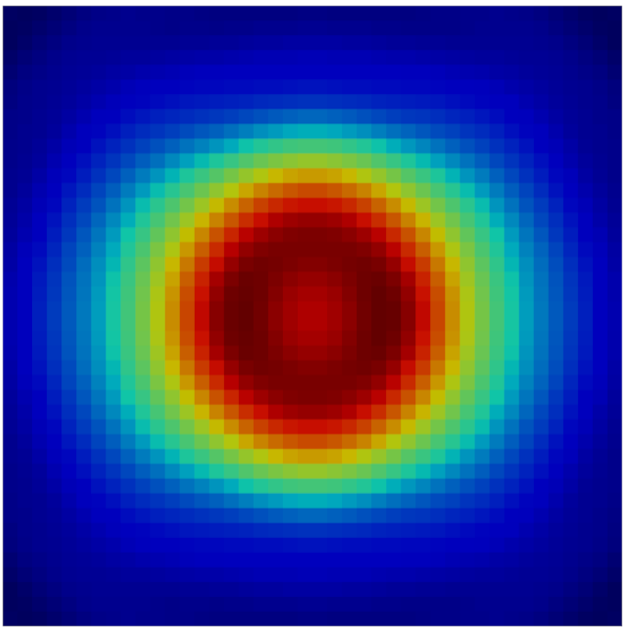


Fig. 7: Dx: 7.5, Size: 780 X 780, mesh-length: 3.93, S1-S2: 220 and 250

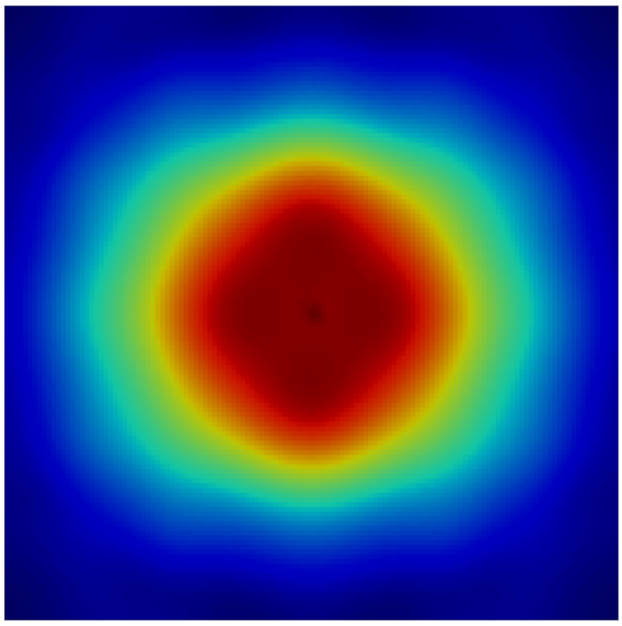


Fig. 8: Dx: 1.9, Size: 190 X 190, mesh-length: 1.27, S1-S2: 200 and 240

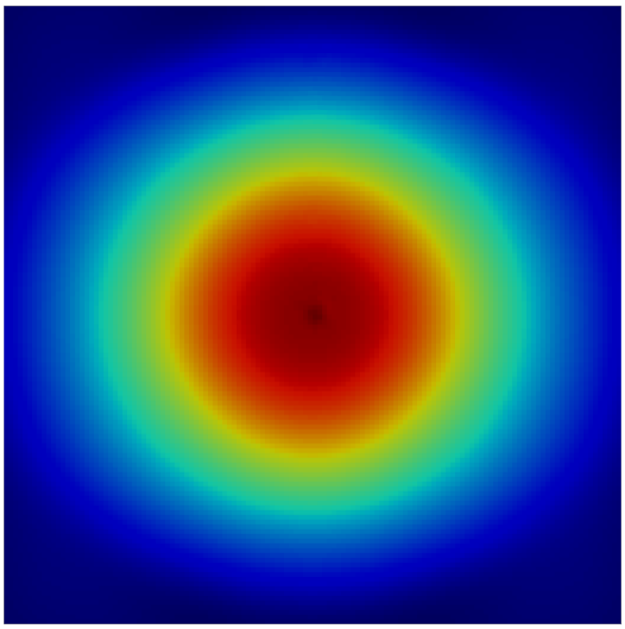


Fig. 9: Dx: 7.5, Size: 760 X 760, mesh-length: 1.27, S1-S2: 290 and 300

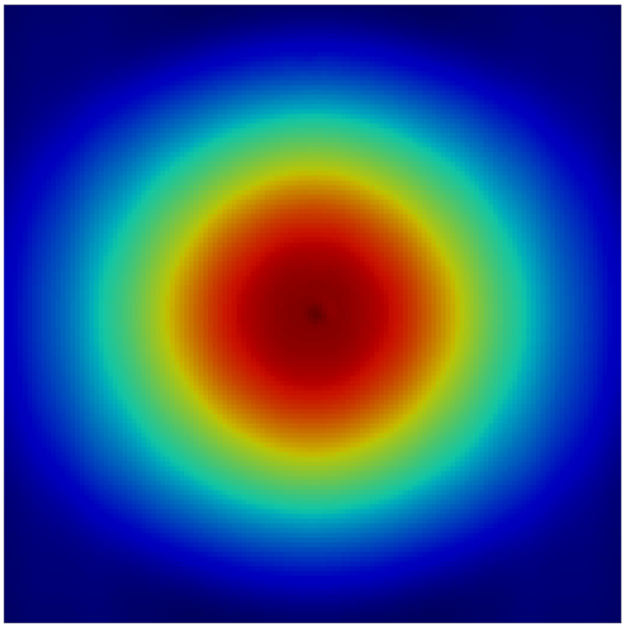


Fig. 10: Dx: 13, Size: 760 X 760, mesh-length: 1.27, S1-S2: 290 and 300

REFERENCES

[1] T. Brown, I. Jeffrey, and P. Mojabi, "Multiplicatively regularized source reconstruction method for phaseless planar near-field antenna measurements," *IEEE Transactions on Antennas and Propagation*, vol. 65, pp. 2020–2031, April 2017.